SHANTU RAJ GOGOI

shanturajgogoi123@gmail.com | +91 9395393823 | linkedin.com/in/shanturajgogoi | Golaghat, Assam, India

PROFESSIONAL SUMMARY

Final-year Mechanical Engineering student with hands-on experience in product design, CAD modeling, manufacturing, and refinery operations, gained through internships at Neodrant Healthcare Solutions, Numaligarh Refinery Limited (NRL), and TRTC MSME Guwahati. Skilled in SolidWorks, ANSYS Workbench, GD&T, Finite Element Analysis (FEA), mechanical design, and manufacturing processes, with a track record of delivering production-ready CAD models and engineering drawings.

CORE COMPETENCIES

Mechanical Design • Product Development • Machine Design • Manufacturing Processes • Equipment Inspection • Preventive Maintenance•EngineeringAnalysis•ProblemSolving•Technical Documentation•Team Collaboration

TECHNICAL SKILLS

Design & Analysis: SolidWorks, ANSYS Workbench, CAD Modeling, GD&T (Geometric Dimensioning & Tolerancing), Finite Element Analysis (FEA), Factor of Safety (FOS), Computational Fluid Dynamics (CFD), Engineering Drawing.
Core Engineering: Fluid Mechanics, Heat Transfer, Mechanical Maintenance & Inspection, Manufacturing Processes, Microsoft Office

PROFESSIONAL EXPERIENCE

Design Engineer Intern *Aug2026–Present*
Neodrant Healthcare Solutions Pvt. Ltd.
- Contribute to product design and manufacturing of electric toothbrush components through 3D CAD modeling and iterative design modifications.

Mechanical Engineering Intern *Jul2026–Aug2026*
Numaligarh Refinery Limited (NRL), Assam
- Studied end-to-end refinery operations from crude oil distillation through down stream processing, gaining hands on exposure to pumps, compressors, heat exchangers, boilers, and refinery safety practices.

CAD Design Intern *Jun2025–Jul2025*
TRTC (Tool Room & Training Centre), MSME, Guwahati
- Designed 15+ mechanical components and assemblies using SolidWorks form an ufacturing applications.
- Prepared 3D models and detailed engineering drawings to support production ready designs.

PROJECTS

Arduino Based Robotic Arm
- Designed and assemble damulti-degree-of-freedom(DOF) robotic arm using Arduino and servomotors, fromCAD modeling through mechanical assembly.

Mini F-22 Raptor RC Drone for Emergency Supply Delivery
- Designed a lightweight, F-22Raptor-inspired RC drone to deliver medicines and essential supplies to flood-affected and remote areas, applying CAD modeling, aerodynamic principles, and mechanical design to optimize flight per formance and payload capacity.

Knuckle Joint Design & Structural Analysis
- Designed a knuckle joint in SolidWorks and validated structural strength in ANSYS under a 10 kN static load, confirming design safety through Factor of Safety analysis.

LEADERSHIP EXPERIENCE

Research & Development Head *2026–Present*
Design Club, BVEC
- Lead CAD-based engineering projects and mentor students in SolidWorks.

Joint Secretary *2025–2026*
Robotronix Club, BVEC
- Organized technical events and coordinated robotics projects.

EDUCATION

Bachelor of Technology in Mechanical Engineering *2023–2027*
Barak Valley Engineering College (BVEC)

CERTIFICATIONS

- Mechanical Design using SolidWorks– TRTC, MSME Guwahati
- ANSYS Workbench Complete Course–Udemy
- Drone Bootcamp– NIT Silchar
- Robotics with IoT & AI Workshop–BVEC